MAST20030 Differential Equations Semester 2 Exam 2017

Differential Equations (University of Melbourne)



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THE UNIVERSITY OF MELBOURNE

Student Number

Semester 2 Assessment, 2017

School of Mathematics and Statistics

MAST20030 Differential Equations

Writing time: 3 hours

Reading time: 15 minutes

This is NOT an open book exam

This paper consists of 4 pages (including this page)

Authorised Materials

- Mobile phones, smart watches and internet or communication devices are forbidden.
- Calculators, tablet devices or computers must not be used.
- One A4 sheet, **handwritten only** on both sides, is permitted.
- No books, printed/photocopied materials or notes are permitted.

Instructions to Students

- You must NOT remove this question paper at the conclusion of the examination.
- You should attempt all questions. There are six questions whose marks are indicated.
- The total number of marks available is 100.

Instructions to Invigilators

- Students must NOT remove this question paper at the conclusion of the examination.

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Question 1 (15 marks)

In this question, we consider the homogeneous third-order linear ODE

$$t^3(1-t)\frac{\mathrm{d}^3y}{\mathrm{d}t^3} + t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2t\frac{\mathrm{d}y}{\mathrm{d}t} + 2y(t) = 0. \tag{1}$$

(a) Show that the functions $y_1(t) = t$ and $y_2(t) = t^2$ are linearly independent solutions. (*3 marks*)

(b) Use the wronskian

$$\mathrm{wr}[y_1, y_2, y_3] = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

to deduce that *any* solution $y_3(t)$ to the ODE in (1), such that $\{y_1, y_2, y_3\}$ is linearly independent, satisfies the following *inhomogeneous* second-order linear ODE:

$$t^2\frac{\mathrm{d}^2y_3}{\mathrm{d}t^2} - 2t\frac{\mathrm{d}y_3}{\mathrm{d}t} + 2y_3(t) = \frac{t-1}{t}. \tag{*5 marks*}$$

(c) Verify that t and t^2 are complementary solutions to this ODE and then use reduction of order to find a solution y_3 such that $\{y_1, y_2, y_3\}$ is linearly independent. (*5 marks*)

(d) Write down the general solution to the ODE in (1). (*2 marks*)

Question 2 (15 marks)

This question concerns the following homogeneous second-order linear ODE:

$$ty''(t) + y'(t) + ty(t) = 0.$$

(a) By trialling a series solution of the form

$$y_1(t) = \sum_{j=0}^{\infty} c_j t^j,$$

show that c_1 must be 0. Write down the formal solution $y_1(t)$ for which $c_0 = 1$. (*6 marks*)

(b) What is the radius of convergence of your formal solution? On which intervals of $\mathbb{R}$ does it converge *uniformly*? (*3 marks*)

(c) By quoting any relevant results from lectures, state whether your formal solution $y_1(t)$ is a genuine solution or not. [*You do not have to show that your solution is genuine.*] (*2 marks*)

(d) Compute the wronskian of $y_1(t)$ with another linearly independent solution $y_2(t)$. Use the result to show that $y_2(t)$ necessarily has a logarithmic singularity at $t = 0$. (*4 marks*)

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Question 3 (15 marks)
Consider the second-order linear IVP

$$4y''(t) + 12y'(t) + 9y(t) = 2\delta_2^+(t), \qquad y(0) = 2, \quad y'(0) = -2.$$

(a) Use Laplace transforms to solve this IVP for $y(t)$. Write your answer as a piecewise-defined function (no Heavisides allowed). (*8 marks*)

(b) Check explicitly if $y(t)$ is continuous at $t = 2$. Is it differentiable there? (*5 marks*)

(c) Sketch a rough plot of $y(t)$ for $0 \leqslant t \leqslant 4$, illustrating x- and y-intercepts as well as your results for the previous part. (*2 marks*)

Question 4 (15 marks)
Consider the following mixed BVP:

$$y''(x) + \lambda y(x) = 0, \qquad y(0) = y'(L) = 0.$$

(a) Show, from first principles, that the eigenvalues and eigenfunctions of this BVP are

$$\lambda_n = \frac{\pi^2(n - \frac{1}{2})^2}{L^2}, \qquad y_n(x) = \sin\frac{\pi(n - \frac{1}{2})x}{L}, \qquad n = 1, 2, 3, \ldots \quad (5 \text{ marks})$$

(b) Fill in the blanks: These eigenfunctions form _____ for a _____ of functions satisfying _____. (*3 marks*)

(c) Use your answer to the previous part to prove that

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{\pi(n - \frac{1}{2})x}{L} \qquad \Rightarrow \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{\pi(n - \frac{1}{2})x}{L}\,\mathrm{d}x. \quad (3 \text{ marks})$$

(d) Show that the b_n when $f(x) = x$ have the form $b_n = \dfrac{2L(-1)^{n-1}}{\pi^2(n - \frac{1}{2})^2}$. (*4 marks*)

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Question 5 (25 marks)

Marlene, Darlene and Sharlene are building a mega-extreme ultimate skipping machine (TM). Their first prototype has two motors spaced π metres apart (they're geeky like that) whose purpose is to control the vertical position and the angle the rope makes at each end. We model the height $u(x,t)$ of the rope using the wave equation and the following motor control instructions:

$$u(0,t) = 0, \qquad \partial_x u(\pi,t) = 1.$$

The ladies set their initial rope configuration so that $u(x,0) = x$ and $\partial_t u(x,0) = -x$.

(a) Write $u(x,t) = u_{\text{lin.}}(x,t) + u_{\text{rest}}(x,t)$, where $u_{\text{lin.}}(x,t)$ is chosen so that $\partial_x^2 u_{\text{lin.}}(x,t) = 0$ and it satisfies the inhomogeneous boundary conditions above. Solve for $u_{\text{lin.}}$ and show that $u_{\text{rest}}(x,t)$ satisfies an IBVP with homogeneous boundary conditions. Write this IBVP out in full. (*5 marks*)

(b) Find a formal solution to this homogeneous IBVP and thus write down a formal solution for $u(x,t)$. (*10 marks*)

(c) What would you have to check in order to prove that this solution was a genuine solution of the wave equation? [*You do not need to demonstrate it!*] (*4 marks*)

(d) Consider the energy functional

$$E[v](t) = \frac{1}{2}\int_0^\pi \left[(\partial_t v(x,t))^2 + c^2(\partial_x v(x,t))^2\right] \mathrm{d}x.$$

Show that $E[v](t) = 0$ for all $t \geqslant 0$, where v is the difference of any two solutions to the original IBVP (with inhomogeneous boundary conditions). Then, explain why this proves that the solution $u(x,t)$ found above is unique. (*6 marks*)

Question 6 (15 marks)

We wish to find the steady state temperature distribution $u(x,y)$ of a thin copper strip of effectively infinite length whose temperature is fixed to $u(x,0) = 0$ along the boundary $y = 0$ and to $u(x,1) = (1+x^2)^{-1}$ along the boundary $y = 1$. Modelling the steady state distribution as a solution of Laplace's equation, compute the steady state solution as follows:

(a) Take the *complex* Fourier transform $\mathscr{F}_x$ of Laplace's equation with respect to x (so y is treated as a constant) and obtain a differential equation in y only. (*2 marks*)

(b) Find the general solution of this differential equation for $(\mathscr{F}_x u)(p,y)$. You should have two unknown functions of p in your answer. (*3 marks*)

(c) Apply $\mathscr{F}_x$ to the two boundary conditions and thereby fix both functions of p. Write down the Fourier transform of the solution $u(x,y)$. (*5 marks*)

(d) Write down the solution $u(x,y)$ as an integral over p. Don't try to evaluate the integral. (*2 marks*)

(e) Your result should be written in terms of complex-valued functions, but the solution must clearly be real. Show that the imaginary part of your result vanishes identically. (*3 marks*)

End of Exam—Total Available Marks = 100

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